Couturier 2020 GBM — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) on sample integrates 10 patient tumours: sample iLISI 1.16→4.43 of 10 (plateau; peaks at θ=10). One connected component, 0 detached islands. Judged on the regenerated embedding + per-neighbourhood iLISI: malignant structure is mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

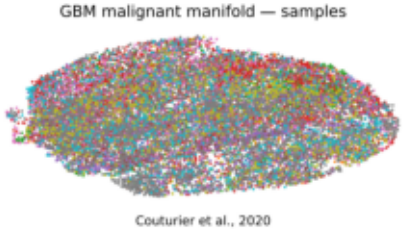
Non-branching: a single Gast/pSom → Crest axis. Gast & pSom co-vary (smoothed r=+0.53), both oppose Crest (Gast↔Crest r=−0.73, pSom↔Crest r=−0.81). Along pseudotime Gast/pSom fall (ρ −0.48 / −0.41), Crest rises to a terminal tip (Crest-high decile mean pseudotime = 0.94).

Gate 3 — cross-repression → NA

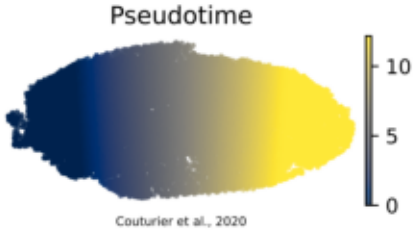
Not applicable: the manifold is non-branching (principal curve, 2 tips / 0 forks), so there is no Gast/pSom-opposed bifurcation to test. Branching-only correlation / modules notebooks are not produced (per spec section 6).

STEP FIGURES

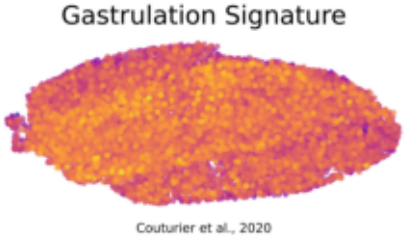
Manifold — samples



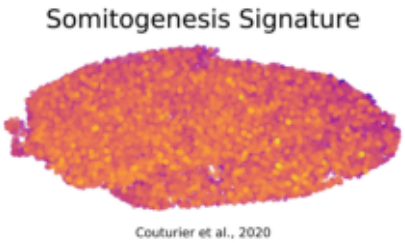
Pseudotime



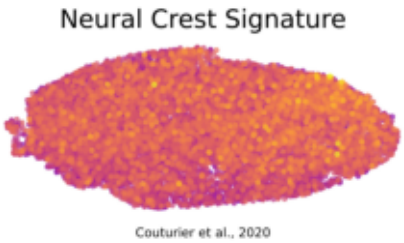
Gastrulation



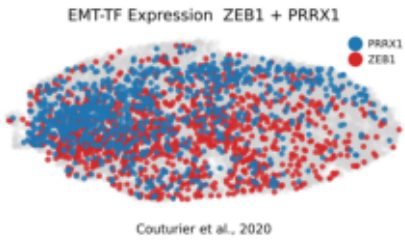
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Glioblastoma (Couturier et al., 2020), 10x — 16,804 malignant cells (10 patient tumours). Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces (non-branching, Linear-pSom/Crest): one continuous gastrulation/somitogenesis → neural-crest axis rooted at the Gast/pSom-high origin, Crest terminal. G1 GO / G2 GO / G3 NA. EMT-TFs ZEB1 + PRRX1 (both terminus-enriched; no root-end factor).

Parameter choices

Sample threshold	≥418 cells (536→327 gap, ratio 1.64); 10/29 samples, 16,804 cells	**Gene selection / PCA**	scFates find_overdispersed = 2,963 genes; PCA 50 comps
Integration	Harmony on sample θ=10 (iLISI 1.16→4.43 plateau)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 0 islands removed
Geometry	Non-branching; scFates curve N=20, λ=0.3, μ=100 (2 tips / 0 forks)	**Root node**	curve tip 0 = Gast/pSom-high, Crest-low (developmental origin)
Module means (raw)	Gast +0.004 · pSom +0.077 · Crest −0.044 (pSom strongest)	**EMT-TFs**	ZEB1 + PRRX1 (both terminus-enriched, distinct families); no root factor